12-Month Roadmap to Become an AI Engineer

Month 1 – Python Programming Fundamentals

- Python syntax, variables, data types
- Loops, conditionals, functions
- File handling and exception handling
- Libraries: NumPy, Pandas
- Project: Simple dataset analysis

Month 2 – OOP & Data Handling

- Object-Oriented Programming (classes, objects, inheritance)
- Data structures in Python
- Tools: Git, GitHub, Jupyter Notebook, VS Code
- Statistics basics: mean, median, variance, standard deviation
- Project: Dataset visualization

Month 3 – Linear Algebra for AI

- Vectors and matrices
- Matrix multiplication and dot product
- Eigenvalues (basic concept)
- NumPy matrix operations
- Project: Implement vector operations in Python

Month 4 – Probability & Statistics

- Probability rules and conditional probability
- Bayes theorem
- Normal and binomial distributions
- Hypothesis testing and confidence intervals
- Project: Statistical analysis of a dataset

Month 5 – Data Analysis & Feature Engineering

- Exploratory Data Analysis (EDA)
- Data cleaning and missing value handling
- Feature scaling and selection
- Tools: Pandas, Matplotlib, Seaborn
- Project: Full EDA on a Kaggle dataset

Month 6 – Machine Learning Fundamentals

- Algorithms: Linear Regression, Logistic Regression
- KNN, Decision Trees, Random Forest, SVM
- Concepts: Bias vs variance, cross-validation
- Library: Scikit-learn
- Project: House price prediction

Month 7 – Advanced Machine Learning

- Gradient Boosting
- XGBoost and LightGBM
- Hyperparameter tuning
- Model evaluation metrics
- Project: Fraud detection or credit risk model

Month 8 – Calculus & Optimization

- Derivatives and partial derivatives
- Gradient descent
- Loss functions
- Project: Implement gradient descent in Python

Month 9 – Deep Learning

- Neural networks and activation functions
- Backpropagation
- CNN (Convolutional Neural Networks)
- Framework: PyTorch
- Project: Image classification model

Month 10 – Natural Language Processing

- Tokenization and word embeddings
- TF-IDF
- Sentiment analysis
- Libraries: NLTK, SpaCy
- Project: Twitter sentiment analysis

Month 11 – Generative AI & LLMs

- Transformer architecture

- Prompt engineering
- Retrieval Augmented Generation (RAG)
- Tools: Hugging Face, LangChain, LlamaIndex
- Project: AI chatbot for document Q&A;

Month 12 – Deployment & MLOps

- FastAPI or Flask
- REST APIs
- Docker basics
- Cloud deployment basics (AWS/GCP)
- Project: Deploy ML model as API